

Racial & Ethnic Disparities in Postoperative Opioid Prescribing Following Spine Surgery

Hybrid ANGLE 2+4 v2 Analysis | N = 1,109 Patients | Comprehensive Research Evaluation Report

1. Research Overview & Background

This study investigates racial and ethnic disparities in postoperative opioid prescribing among patients undergoing spine surgery. Using the HYBRID ANGLE 2+4 v2 dataset — an electronic health record (EHR) and patient-reported outcome (PRO) linkage — the research examines whether Non-Hispanic Black (NHB) and Hispanic/Latino (HL) patients receive systematically different amounts of postoperative opioid medications compared to Non-Hispanic White (NHW) patients, and whether social determinants of health (SDOH) — including healthcare access barriers, discrimination, pain levels, and socioeconomic factors — explain any observed disparities.

The study is situated within a broader literature documenting that racial minorities in the United States consistently receive fewer opioid prescriptions than White patients despite reporting equivalent or greater pain — a phenomenon sometimes called 'opioid underprescribing' or 'pain undertreatment' in minority populations. Simultaneously, minority populations face documented barriers to healthcare access and experience higher rates of discrimination in clinical settings. This study attempts to disentangle these competing forces using a sequential regression design.

Content Gap Being Addressed

Gap	Description
Underrepresentation in opioid research	Prior studies on postoperative opioids rarely include adequate minority samples or link EHR data with patient-reported SDOH measures simultaneously.
Mechanistic ambiguity	It is unknown whether disparities in prescribing are explained by legitimate clinical factors (pain level, comorbidities), barriers to care, discrimination, or unmeasured social variables.
Preoperative opioid status interaction	The moderating role of preoperative chronic opioid use on the race-prescribing relationship has not been systematically studied.
Multidimensional SDOH integration	Few studies jointly model financial barriers, logistic barriers, healthcare discrimination, everyday discrimination, food insecurity, social support, and stress as mediators/confounders.

2. Study Groups & Cohort Characteristics

The full analytic cohort comprised **1,109 patients** with spine disorders drawn from a single health system. Patients were grouped by race/ethnicity as reported in the EHR and/or self-identified via survey.

Group	N	%	Mean Age	% Female	% Pre-op Chronic Opioid	% Any Postop Opioid
Non-Hispanic White (NHW)	818	73.8%	51.6 ± 14.5	56.6%	29.8%	57.8%
Non-Hispanic Black (NHB)	123	11.1%	51.5 ± 10.1	61.8%	33.3%	52.8%
Hispanic/Latino (HL)	77	6.9%	47.8 ± 12.5	64.9%	36.4%	53.2%
Other/Unknown	82	7.4%	—	—	—	—
Non-Hispanic Asian	9	0.8%	—	—	—	—
MINORITY (NHB + HL)	200	18.0%	50.1 ± 11.2	63.0%	34.5%	53.0%

Table 1. Cohort composition by race/ethnicity group.

The analytic design uses both a **3-group comparison** (NHW vs. NHB vs. HL) and a **2-group comparison** (NHW vs. combined Minority). The 2-group approach improves statistical power for the minority sample (N=200), though it obscures NHB vs. HL differences that are visible in the 3-group model.

Key Baseline Differences

Variable	NHW	NHB	HL	Minority	p-value
Mean 7-day pain score	3.5 ± 2.6	5.9 ± 2.7	6.4 ± 2.5	6.1 ± 2.6	<0.0001 ✓
Depression (%)	43.6%	51.2%	70.1%	58.5%	—
Chronic pain (%)	30.8%	34.1%	19.5%	28.5%	—
Any access barrier (%)	30.0%	15.4%	13.0%	14.5%	<0.0001 ✓
Any HC discrimination (%)	42.5%	26.8%	14.3%	22.0%	0.457 ✗
Postop Rx count	6.3 ± 19.9	13.9 ± 112.7	4.2 ± 9.3	10.2 ± 88.6	0.345 ✗

Table 2. Between-group differences (Kruskal-Wallis, 3-group). ✓ = significant, ✗ = not significant.

3. Opioid Outcomes & Results

Three primary postoperative opioid outcomes were analyzed within 90 days of surgery:

Outcome	Full Cohort	NHW	Minority	Significance
Any postop opioid	55.8% (619/1109)	57.8%	53.0%	NS (p=0.30 race-only)
Mean Rx count	7.0 ± 42.1	6.3 ± 19.9	10.2 ± 88.6	NHB elevated (p<0.001)
Prolonged use (>30d)	26.8%	26.4%	28.5%	NS
Preop chronic opioid	30.7%	29.8%	34.5%	—

Table 3. Primary opioid outcomes by group.

Key finding: While simple binary opioid receipt (yes/no) does not differ significantly by race, the *volume* of prescriptions (Rx count) shows a striking racial pattern. Non-Hispanic Black patients have a mean Rx count nearly 2.2× that of NHW patients in the unadjusted model (IRR=2.19, p<0.001). This counterintuitive finding — higher Rx count for NHB patients despite reporting more barriers and discrimination — warrants careful interpretation and is addressed in the regression analyses below.

Stratification by Preoperative Opioid Status

A critical moderator emerged when the cohort was split by preoperative chronic opioid use:

Subgroup	NHW (n)	NHW Opioid Rate	NHW Rx Count	Minority (n)	Minority Rate	Minority Rx Count	Minority IRR
Non-chronic (opioid-naïve + acute)	574	52.1%	4.2 ± 16.4	131	39.7%	11.9 ± 109.1	1.36*
Chronic preop opioid users	244	71.3%	11.3 ± 25.8	69	78.3%	6.8 ± 13.9	0.64*

Table 4. Stratified results. *p<0.01. IRR = Incident Rate Ratio (NegBin regression).

Among **opioid-naïve/acute patients**, minority patients receive *more* prescriptions than NHW (IRR=1.36). Among **chronic preop opioid users**, the pattern reverses: minority patients receive *fewer* prescriptions than NHW (IRR=0.64). The interaction term (minority × chronic) is highly significant (IRR=0.44, p<0.001), confirming this moderation.

4. Sequential Regression Analysis

Sequential (hierarchical) regressions were run to evaluate how different SDOH blocks attenuate or amplify racial/ethnic gaps. Two outcomes were modeled: (A) any postop opioid (logistic) and (B) Rx count (negative binomial). Models were labeled A through F, each adding a covariate block:

Model	Covariates Added	Purpose
A	Race/ethnicity only	Unadjusted baseline disparity
B	+ Age, sex, preop chronic opioid, comorbidity count	Clinical confounders
C	+ 7-day pain score (PROMIS)	Pain-mediated pathway
D	+ Financial and logistic barriers (HCAU)	Access barrier pathway
E	+ Healthcare discrimination (HC Discrim)	Discrimination pathway
F	+ Food insecurity, social support, stress	Social factors (FAILED for logistic)

Table 5. Sequential regression model blocks.

4A. Outcome: Any Postop Opioid (Binary Logistic)

No significant racial/ethnic differences in binary opioid receipt were found in any model. The race coefficients were not significant across all blocks ($p>0.29$ for NHB; $p>0.44$ for HL at baseline). Key drivers of any opioid receipt were **preoperative chronic opioid use** (OR=4.69, $p<0.001$) and **comorbidity count** (OR=0.15, $p<0.001$). The attenuation pattern for NHB showed 60–80% attenuation after adding demographics, but the HL coefficient reversed direction (β went from -0.185 to $+0.097$), suggesting suppression or confounding rather than mediation.

Model	NHB OR (95%CI)	NHB p	HL OR (95%CI)	HL p
A – Race only	0.82 (0.56–1.20)	0.299	0.83 (0.52–1.33)	0.438
B – +Demographics	0.92 (0.60–1.41)	0.712	1.04 (0.62–1.76)	0.877
C – +Pain	0.95 (0.61–1.47)	0.813	1.07 (0.63–1.84)	0.797
D – +Barriers	0.96 (0.62–1.49)	0.857	1.10 (0.64–1.90)	0.726
E – +Discrimination	0.95 (0.61–1.48)	0.811	1.07 (0.62–1.85)	0.802
F – +Social	FAILED	—	FAILED	—

Table 6. Logistic regression — any postop opioid.

4B. Outcome: Prescription Count (Negative Binomial — 3-Group)

The NHB–NHW gap in Rx count was significant and *grew* after adjusting for demographics and pain (IRR increased from 2.19 to 3.23), indicating that NHB patients receive more prescriptions than their clinical profile would predict. Healthcare discrimination (Model E) partially attenuated this (hc_discrim IRR=0.73), while logistic barriers significantly increased Rx count for all (IRR=1.54–1.61). The social factors block (Model F) attenuated NHB disparity substantially (IRR dropped to 1.42, attenuation=55.4%), suggesting unmeasured social confounders are at play.

Hispanic/Latino patients showed a protective pattern: significantly *fewer* prescriptions than NHW (IRR=0.67 unadjusted), persisting through all models.

Model	NHB IRR (95%CI)	p	HL IRR (95%CI)	p
A – Race only	2.19 (1.80–2.67)	<0.001*	0.67 (0.52–0.87)	0.002*
B – +Demographics	2.62 (2.14–3.21)	<0.001*	0.66 (0.50–0.85)	0.002*
C – +Pain	3.23 (2.62–3.98)	<0.001*	0.73 (0.56–0.95)	0.022*
D – +Barriers	3.16 (2.56–3.90)	<0.001*	0.76 (0.58–0.99)	0.046*
E – +Discrimination	3.21 (2.60–3.97)	<0.001*	0.72 (0.54–0.94)	0.017*
F – +Social	1.42 (1.15–1.75)	0.001*	0.68 (0.51–0.90)	0.007*

Table 7. Negative binomial — Rx count, 3-group comparison. *p<0.05.

4C. Outcome: Prescription Count (NHW vs. Minority — 2-Group)

The 2-group analysis revealed an unexpected and unstable attenuation pattern. Adding pain (Model C) caused the minority coefficient to flip from positive (IRR=1.82) to null/negative (IRR=0.90, $p=0.25$). This sign flip — a hallmark of multicollinearity or suppressor effects — indicates that **pain and comorbidity count are jointly collinear with minority status**, producing paradoxical coefficient behavior. Adding discrimination (Model E) then caused the minority IRR to jump back up to 2.23 (–69.3% 'attenuation', meaning amplification), suggesting HC discrimination acts as a suppressor or that the model is overfit.

Model	Minority IRR (95%CI)	p	Attenuation vs. A
A – Minority only	1.61 (1.36–1.89)	<0.001**	0.0% (baseline)
B – +Demographics	1.82 (1.54–2.15)	<0.001**	–26.3% (amplified)
C – +Pain	0.90 (0.76–1.08)	0.252	121.8% (over-attenuated)
D – +Barriers	0.96 (0.81–1.15)	0.678	108.0% (sign flip)
E – +Discrimination	2.23 (1.86–2.67)	<0.001**	–69.3% (amplified again)

Table 8. Negative binomial — Rx count, 2-group (NHW vs. Minority). ** $p<0.001$.

5. Interpretation of Results

NHB patients receive more prescriptions per episode (unadjusted)	This finding is counterintuitive given the literature on minority underprescribing. However, it may reflect that NHB patients have more complex, prolonged spine conditions, more severe postoperative pain, or that providers are compensating for undertreated preoperative pain. The NHB Rx count mean is inflated by extreme outliers ($SD = 112.7$), suggesting a bimodal distribution — most NHB patients receive few prescriptions, while a small subset receives very many.
Demographics amplify rather than attenuate the NHB-Rx gap	Adding age, sex, comorbidities, and preop opioid use made the NHB coefficient larger ($2.19 \rightarrow 3.23$). This is 'negative mediation' or suppression: after accounting for clinical complexity, NHB patients are even more likely to receive high Rx counts than their clinical profile predicts. This could mean that clinical factors partially suppress (hide) the true disparity, or that unmeasured confounders correlated with both race and these covariates are distorting estimates.
Hispanic/Latino patients consistently receive fewer prescriptions	HL patients have IRRs of 0.67–0.76 across all models — a robust protective (or discriminatory?) pattern. HL patients also report the highest pain scores (6.4 vs. 3.5 for NHW) but the fewest prescriptions. This is consistent with undertreated pain in HL populations and may reflect language barriers, immigration-related fear, or provider stereotyping.
Healthcare discrimination partially explains the NHB elevation	In Model E (3-group), adding HC discrimination attenuated the NHB IRR slightly and the <code>hc_discrim</code> term was significant (IRR=0.73 for the score itself). Patients who reported more discrimination received fewer prescriptions — consistent with a discrimination-mediated undertreatment pathway.

Preop opioid status is a critical moderator	The interaction (minority × chronic = IRR 0.44) means the minority-opioid relationship is fundamentally different depending on prior opioid exposure. Chronic preop opioid minority patients receive fewer prescriptions postoperatively (possible tapering, stigma, or formulary concerns), while opioid-naïve minority patients receive more. Failing to account for this interaction masks both phenomena.
Financial barrier–minority interaction (IRR = 0.19)	The interaction between minority status and financial barriers is dramatic (IRR=0.19, p<0.001). Minority patients with financial barriers receive far fewer prescriptions than expected — consistent with combined structural disadvantage at the intersection of race and economic insecurity.

6. Analyses That Failed — Causes & Remedies

Several variables or model components produced errors or unreliable results. These are documented below with root causes and recommended fixes.

■ Model F (Logistic) — Singular Matrix

CAUSE: The social factors block (food insecurity, social support, stress) caused a singular/near-singular matrix in the logistic model. This occurs when predictors are perfectly collinear, when a variable has near-zero variance, or when sample size is insufficient relative to the number of cells. Social support had only N=361 respondents (32.6%), and perceived stress N=516 (46.5%) — meaning the social block model runs on a dramatically reduced subset, causing sparse cells and collinearity with race/ethnicity indicators.

✓ **RECOMMENDED FIX:** FIX: (1) Use multiple imputation to handle missing SDOH data rather than complete-case analysis. (2) Test social variables one at a time. (3) Use regularization (Lasso/Ridge logistic). (4) Consider a two-stage model: predict binary opioid receipt in Stage 1, then Rx count among users.

■ PROMIS Variables (fatigue, general health, physical health, mental health, emotional problems, everyday activities) — Parse Failure

CAUSE: All six PROMIS/PRO variables generated 'could not parse answers' warnings and were coded as 0% complete. The values are stored as free-text strings (e.g., 'Average Fatigue 7 Days: Mild') rather than numeric scores. The parsing function expected numeric Likert values but received labeled text. A typo was also detected ('Excllent' for 'Excellent' in mental health), indicating inconsistent data entry.

✓ **RECOMMENDED FIX:** FIX: (1) Recode string responses to ordinal integers (None=0, Mild=1, Moderate=2, Severe=3, Very Severe=4). (2) Implement fuzzy matching to handle typos. (3) Re-run all models including these 6 PROMIS dimensions as mediators. This is a major data loss — these outcomes likely have significant explanatory power.

■ Everyday Discrimination — Zero Variance (Skipped)

CAUSE: The everyday discrimination score had mean=0.00 and any_everyday_discrimination=0, meaning no patient endorsed everyday discrimination. This is implausible in a diverse cohort and suggests a survey administration or coding error — possibly the scale was not administered, incorrectly scored, or the response options were not matched to the scoring algorithm.

✓ **RECOMMENDED FIX:** FIX: (1) Audit the everyday discrimination survey instrument and scoring code. (2) Compare with distribution from source datasets. (3) If genuine zero endorsement, report as a limitation. (4) Consider separating healthcare vs. everyday discrimination instruments.

■ Social Support — Low Variance / NaN IRR

CAUSE: In Model F NegBin, low_social_support showed IRR=1.00 and p=NaN. This indicates the variable has no discriminating power — likely because it was defined as a binary cutoff on a mean=3.00 variable with insufficient spread, or because social support has perfect correlation with another covariate in the model.

✓ **RECOMMENDED FIX:** FIX: (1) Use continuous social support score. (2) Check distribution for ceiling/floor effects. (3) If binary is needed, select a theoretically meaningful threshold (e.g., below the 25th percentile).

■ 2-Group NHW vs. Minority — Sign Flips and Instability

CAUSE: The minority coefficient oscillated wildly across models (IRR: 1.61 → 1.82 → 0.90 → 0.96 → 2.23). This is a classic multicollinearity/suppressor symptom. Pain scores are much higher in minority patients AND correlated with Rx count, creating a suppressor effect when pain is added. Comorbidity count also changed direction (IRR: 2.16 in B → 0.74 in C), confirming model instability.

✓ **RECOMMENDED FIX:** FIX: (1) Test for multicollinearity (VIF > 10 = problem). (2) Use a causal directed acyclic graph (DAG) to determine whether pain should be treated as a confounder or mediator. If pain is ON the causal path from race to opioid prescribing, conditioning on it introduces collider bias. (3) Consider mediation analysis (Baron-Kenny or structural equation modeling) instead of sequential regression.

■ HCAU Survey Missingness (36.2% Non-Response)

CAUSE: Only 707/1109 (63.8%) patients completed the HCAU survey. Barriers data are therefore missing for 402 patients. This creates selection bias — patients who completed surveys may differ systematically from non-completers (e.g., healthier, more engaged, more English-speaking).

✓ **RECOMMENDED FIX: FIX: (1) Multiple imputation by chained equations (MICE). (2) Analyze predictors of survey completion and apply inverse probability weighting (IPW). (3) Report sensitivity analyses with complete-case vs. imputed datasets.**

7. Recommendations to Improve the Analysis & Research

7.1 Data & Measurement Improvements

- Parse all PROMIS variables (fatigue, general health, physical/mental health, emotional problems, everyday activities) — these represent a substantial unanalyzed predictor block. Standardize response coding with a lookup table and fuzzy matching for typos.
- Audit the everyday discrimination scale. A mean of 0.00 across 523 respondents is almost certainly a data error. This variable is theoretically central to the research question.
- Apply multiple imputation (MICE) for SDOH variables with >5% missingness: social support (67.4% missing), stress (53.5%), HCAU barriers (36.2%), HC discrimination (45.7%). Report both complete-case and imputed results.
- Collect surgery type as a covariate — cervical vs. lumbar vs. thoracic spine procedures have fundamentally different opioid needs and should be stratified or controlled.
- Validate and harmonize the 'Rx count' variable. The NHB SD of 112.7 (vs. mean 13.9) indicates extreme outliers. Investigate whether this reflects data errors, readmissions, or legitimate clinical variation. Consider Winsorizing at the 99th percentile or using log-transformed counts.

7.2 Analytical Improvements

- Draw a Directed Acyclic Graph (DAG) before modeling. Determine whether pain and comorbidities are confounders (should be controlled) or mediators (should not be controlled in total effect estimates). Currently, adding pain appears to introduce collider bias in the 2-group model.
- Replace sequential regression with formal causal mediation analysis (e.g., the R 'mediation' or 'PROCESS' macro) to quantify direct and indirect effects of race on opioid prescribing through pain, barriers, and discrimination pathways.
- Address multicollinearity in the 2-group NegBin models. Compute variance inflation factors (VIF) for all predictors and remove or combine variables with $VIF > 10$.
- Use a zero-inflated negative binomial (ZINB) model rather than standard NegBin, given the large proportion of zero-count patients among some groups. Standard NegBin may underfit the excess zeros.
- Separate the analysis of (a) whether a prescription is given from (b) how many prescriptions are given using a two-part Hurdle model. This cleanly distinguishes access-to-any-opioid disparities from quantity-of-opioid disparities.
- Run sensitivity analyses excluding the extreme NHB Rx count outliers to determine whether findings are robust or driven by a handful of patients.
- Include hospital/provider fixed effects or multi-level (mixed effects) models to account for provider-level prescribing variation, which may confound race effects if providers are racially non-randomly distributed across patients.

7.3 Sample & Design Improvements

- Increase minority sample size. With NHB $N=123$ and HL $N=77$, the study is underpowered for the 3-group analysis and for interaction tests. A minimum of $N=200$ per group is generally recommended for regression with 10+ covariates.
- Consider a multi-site design to improve generalizability and increase the minority sample while also allowing for random effects by site.
- Collect prospective qualitative data (patient interviews, chart review) on the subset of NHB patients with extreme Rx counts to understand the clinical narrative behind the outliers.

- Add prescriber-level data (specialty, years of experience, training program) as covariates, since prescribing behavior varies enormously by provider and could explain much of the race-opioid relationship.
- Track inpatient vs. outpatient prescriptions separately. The 14% inpatient encounter rate may confound analyses if inpatient stays lead to qualitatively different prescribing pathways.

7.4 Reporting Improvements

- Report all missing data proportions in Table 1 and use a missing-data flow diagram (CONSORT-style).
- Present mediation analysis results using forest plots showing direct, indirect, and total effects by racial/ethnic group.
- Disaggregate the Minority category in all tables — NHB and HL show opposite patterns for Rx count (2.19× vs. 0.67×) and combining them obscures clinically meaningful differences.
- Add a supplemental appendix with model diagnostics: residual plots, QQ-plots, Cook's distance for influential observations, and overdispersion tests for the NegBin models.
- Report all attenuation coefficients with bootstrap confidence intervals to assess whether attenuation is statistically significant rather than just descriptive.

8. Executive Summary Dashboard

The table below consolidates key findings, data quality issues, and priority action items.

Domain	Finding / Status	Priority Action
Binary opioid receipt	No racial disparity (NS across all models)	Low — de-emphasize this outcome
Rx count — NHB	NHB receives 2–3× more Rxs (significant); attenuated by social factors	HIGH — use ZINB + mediation analysis
Rx count — HL	HL receives 30–40% fewer Rxs despite highest pain scores	HIGH — investigate undertreated pain
Preop opioid interaction	Opposite effects in chronic vs. non-chronic patients ($p<0.001$)	Must report stratified results
PROMIS variables	6 variables FAILED to parse — 0% usable	CRITICAL — fix parsing code immediately
Everyday discrimination	Zero variance — likely data error	CRITICAL — audit data source
Social support / stress	32.6% / 46.5% completeness; NaN in model	Impute or remove from main models
Model F logistic	Singular matrix failure	Use regularization or run separately
2-Group sign flips	Minority IRR unstable across models (1.61→0.90→2.23)	Build DAG; use mediation analysis
HCAU barriers	36.2% missing; financial barrier NS in all models	Apply MICE imputation; IPW
Sample size	NHB=123, HL=77 — underpowered for 3-group	Recruit additional sites

Overall Assessment

This research addresses a clinically important and socially significant question with a well-constructed dataset linking EHR and patient-reported outcomes. The core finding — that Non-Hispanic Black patients receive more opioid prescriptions but Hispanic/Latino patients receive fewer, despite both groups reporting greater pain than White patients — challenges simple narratives of uniform minority underprescribing and points to heterogeneous, group-specific mechanisms.

However, the analysis is currently limited by: (1) six PROMIS variables that failed to parse and are entirely missing from all models; (2) a probable error in the everyday discrimination data; (3) model instability in the 2-group NegBin due to potential mediator conditioning; and (4) small minority subgroup sizes. Addressing these issues — particularly parsing the PROMIS data and applying formal causal mediation analysis with a DAG — would substantially strengthen the study's conclusions and its potential for publication in a high-impact clinical or health disparities journal.